bag-of-words.[1] Recent work of (Juneja et al., 2022) found that partitioning the 100 fine-tuned BERTs on the basis of their HANS-LO performance is essentially equivalent to partitioning them on the basis of *mode connectivity*: that is, the generalizing and heuristic models lie in separate loss-landscape basins. We refer the interested reader to (Juneja et al., 2022) for further details.

But at what layer(s) do the hidden features of the generalizing and heuristic models diverge? For example, are these two subpopulations of models distinct because their earlier layers are different or because their later layers are different? Or both? Neither analysis of OOD performance nor mode connectivity analysis can provide an answer, but both stitching and CKA reveal that the difference between the features of the generalizing and heuristic models is concentrated in later layers of the BERT encoder. It is worth highlighting that conceptualizing and building relevant datasets for distribution shifts is often expensive and time consuming. That identity stitching and CKA *using in-distribution MNLI data alone* differentiate between heuristic and generalizing behavior on HANS-LO suggests that they are useful tools for model error analysis and debugging.

Figure 2a displays performance of pairs of BERT models stitched with the *identity function* $\varphi = \mathrm{id}$, i.e. models of the form $g_{>m} \circ f_{\leq l}$ on the MNLI finetuning task.[2] We see that at early layers of the models identity stitching incurs almost no penalty, but that stitching at later layers (when the fraction of layers occupied by at the bottom model exceeds 75 %) stitching *between* generalizing and heuristic models incurs significant accuracy drop (high stitching penalty), whereas stitching *within* the generalizing and heuristic groups incurs almost no penalty.

A similar picture emerges in fig. 2b, where we plot CKA values between features of fine-tuned BERTs on the MNLI dataset. At early layers, CKA is insensitive to the generalizing or heuristic nature of model A and B. At later layers (in the same range where we saw identity stitching penalties appear), the CKA measure *between* generalizing and heuristic models significantly exceeds its value *within* the generalizing and heuristic groups.

Together, the results of fig. 2 paint the following picture: the NLI models of (McCoy et al., 2020) seem to all build up generally useful features in their early layers, and only decide to memorize (i.e., use a lexical overlap heuristic) or generalize (use syntactic features) in their final layers.

5 Representation dissimilarity in scaling families

Enormous quantities of research and engineering energy have been devoted to design, training and evaluation of *scaling families* of language models. By a "scaling family" we mean a collection of neural network architectures with similar components, but with variable width and/or depth resulting in a sequence of models with increasing size (as measured by number of parameters). Pioneering work in computer vision used CKA to discover interesting properties of hidden features that vary with network width and depth (Nguyen et al., 2021), but to the best of our knowledge no similar studies have appeared in the natural language domain

We take a first step in this direction by computing CKA measures of features within and between the models of the Pythia family (Biderman et al., 2023), up to the second-largest model with 6.9 billion parameters. In all experiments we use the "`deduped`" models (trained on the deduplicated version of the Pile (Gao et al., 2020)). In the case of inter-model CKA measurements on `pythia-6.9b` we see in fig. 3 (bottom right) that similarity between layers l and m gradually decreases as $|m-l|$ decreases. We also see a pattern reminiscent of the "block structure" analyzed in (Nguyen et al., 2021), where CKA values are relatively high when comparing two layers *within* one of the following three groups: **(i)** early layers (the first 3), **(ii)** late layers (in this case only the final layer) and **(iii)** the remaining intermediate layers, while CKA values are relatively low when comparing two layers *between* these three groups.

Using CKA to compare features *between* `pythia-{1b,1.4b,2.8b}` and `pythia-6.9b` we observe high similarity between features at the be-

[1] By "relatively successful" we mean "Achieving up to 50% accuracy on an adversarially-designed test set." While variable OOD performance with respect to fine-tuning seed was also seen on other subsets of HANS, we follow (McCoy et al., 2020; Juneja et al., 2022) in focusing on HANS-LO where such variance was most pronounced.

[2] The motivation for stitching with a constant identity function comes from evidence that stitching-type algorithms perform symmetry correction (accounting for the fact that features of model A could differ from those of model B by an architecture symmetry e.g. permuting neurons, see e.g. (Ainsworth et al., 2023)), but that networks obtained from multiple fine-tuning runs from a fixed pretrained model seem to not differ by such symmetries (see for example (Wortsman et al., 2022)).

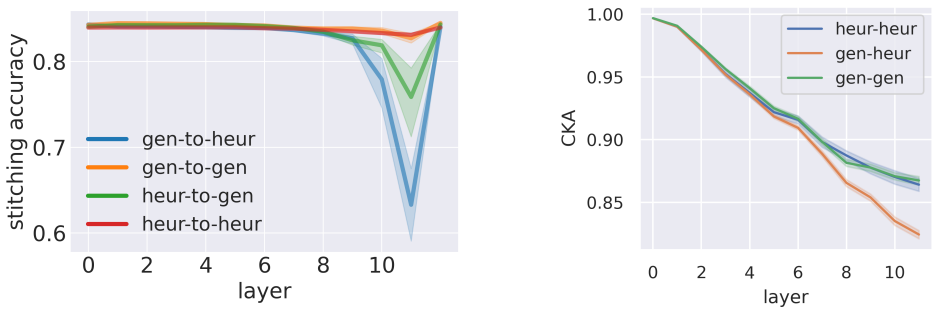

Figure 2: **Left:** Identity stitching on the MNLI dataset between pairs of the top 10 ("generalizing") and bottom 10 ("heuristic") performing models of (McCoy et al., 2020) on the lexical overlap subset of HANS (an out-of-distribution NLI dataset). **Right:** Corresponding CKA values. **Both:** Confidence intervals are obtained by evaluating on all distinct pairs of generalizing and heuristic models, for a total of $2 \cdot \binom{10}{2} + 10^2 = 190$ model comparisons.

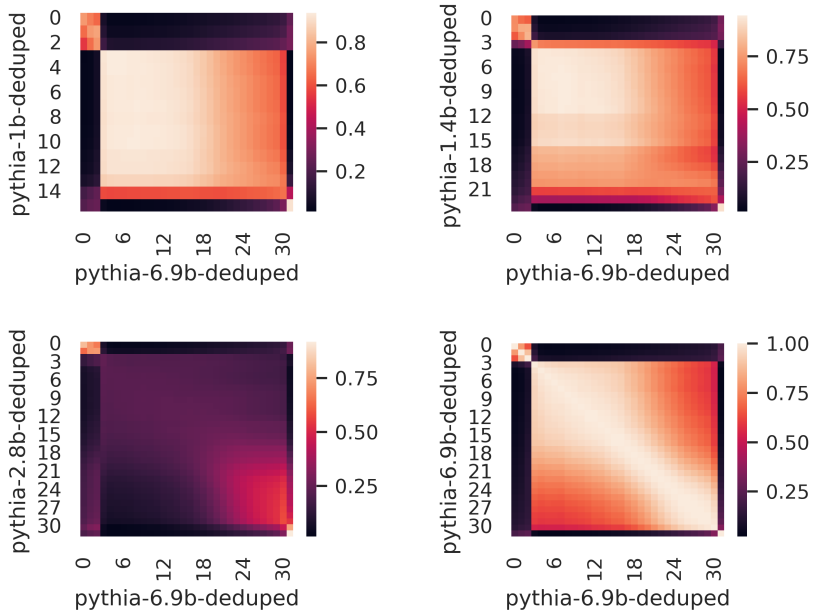

Figure 3: CKA between {`pythia-1b`,`pythia-1.4b`,`pythia-2.8b`,`pythia-6.9b`} and `pythia-6.9b` evaluated on the Pile dataset (Gao et al., 2020) (higher means more similar).

ginning of the model. A plausible reason for this early layer similarity is the common task of "detokenization," (Elhage et al., 2022) where early neurons in models may change unnatural tokenizations to more useful representations, for example responding strongly to compound words (e.g., `birthday | party`). We also continue to see "block structure" even when comparing between two models, and in the case of the pairs (`pythia-1b`, `pythia-6.9b`) and (`pythia-1.4b`, `pythia-6.9b`) substantial inter-model feature similarity in intermediate layers.

The low CKA values obtained when comparing intermediate layers of (`pythia-2.8b`, `pythia-6.9b`) break this trend – in fact, as illustrated in fig. 9, `pythia-2.8b` exhibits such trend-breaking intermediate layer feature dissimilarity with every Pythia model with 1 billion or more parameters.[3] Upon close inspection, we see that while the `pythia-`{`1b,1.4b,6.9b`} models all have a early layer "block" consisting of 3 layers with relatively high CKA similarity, in `pythia-2.8b` this block consists of only 2 layers, suggesting the features of `pythia-2.8b` diverge from the rest of the family very early in the model. In table 1 we point out that some aspects of `pythia-2.8b`'s architecture are inconsistent with the general scaling trend of the Pythia family as a whole.

[3]Except itself, of course.

Limitations

In the current work we examine relatively small language models. Larger models have qualitatively different features, and it is not obvious if our experiments on the differences in layer expressiveness between GeLU and SoLU models will scale to significantly larger models. While we show that identity stitching and CKA distinguish the heuristic and generalizing BERT models, we did not attempt to use stitching/CKA to automatically cluster the models (as was done with mode connectivity in (Juneja et al., 2022)).

Ethics Statement

In this paper we present new applications of representation analysis to language model hidden features. Large language models have the potential to impact human society in ways that we are only beginning to glimpse. A deeper understanding of the features they learn could be exploited for both positive and negative effect. Positive use cases include methods for enhancing language models to be more safe, generalizable and robust (one such approach hinted at in the experiments of section 4), methods for explaining the decisions of language models and identifying model components responsible for unwanted behavior. Unfortunately, these tools can at times be repurposed to do harm, for example extracting information from model training data and inducing specific undesirable model predictions.

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